# IGCSE 0607 Practice Questions — E6.2 — Combined Questions & Markscheme

## Area and perimeter

**Total Questions:** 9 — **Total Marks:** 60 Marks

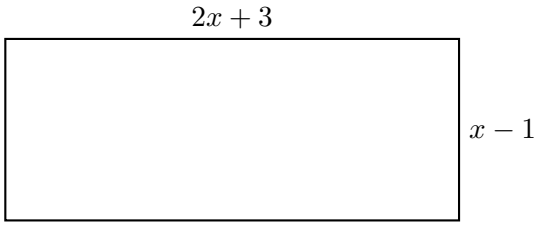
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**1** The diagram shows a rectangle.
All measurements are in centimetres.



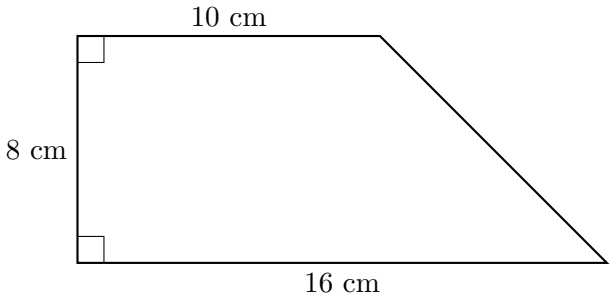

**(a)** The perimeter of the rectangle is 34 cm.
Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Calculate the area of the rectangle.

..... $cm^2$ **[2]**

**2** The diagram shows a right-angled trapezium.



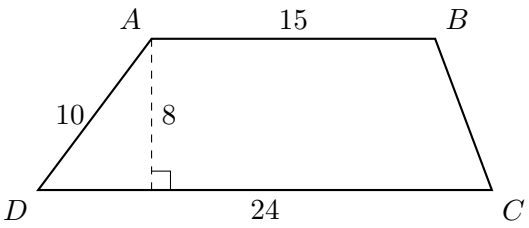

**(a)** Calculate the area of the trapezium.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the trapezium.

..... cm **[3]**

**3** The diagram shows a trapezium $ABCD$ with $AB$ parallel to $DC$.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



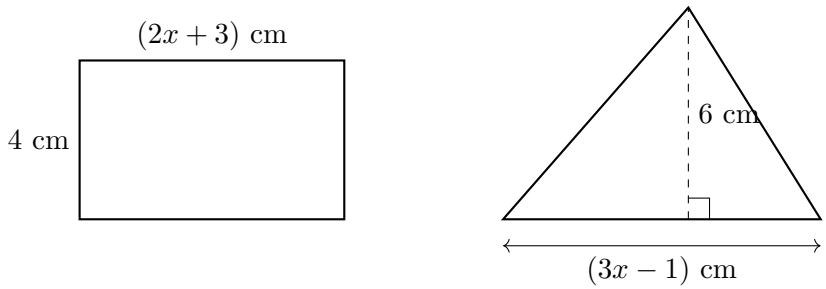

**(a)** Calculate the area of the trapezium.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the trapezium.

..... cm **[4]**

**4** The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



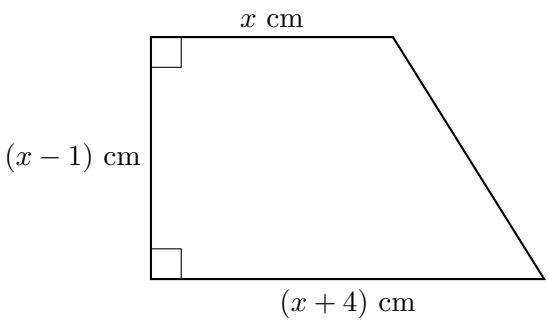

**(a)** Form an equation in $x$ and solve it to find the value of $x$.

$x =$ ..... **[4]**

**(b)** Work out the perimeter of the rectangle.

..... cm **[2]**

**5** The diagram shows a right-angled trapezium with parallel sides of length $x$ cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm$^2$.




**(a)** Show that $x^2 + x - 42 = 0$.

**[3]**

**(b)** Solve the equation $x^2 + x - 42 = 0$ to find the value of $x$.

$x =$ ..... **[2]**

**(c)** Calculate the perimeter of the trapezium.

..... cm **[4]**

**6** A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be $y$ cm.

**(a)** Write down an expression, in terms of $y$, for the side length of the square.

..... cm **[2]**

**(b)** The total area enclosed by the square and the rectangle is 113 $\text{cm}^2$.
Show that $17y^2 - 180y + 448 = 0$.

**[3]**

**(c)** Solve the equation.

$$17y^2 - 180y + 448 = 0$$

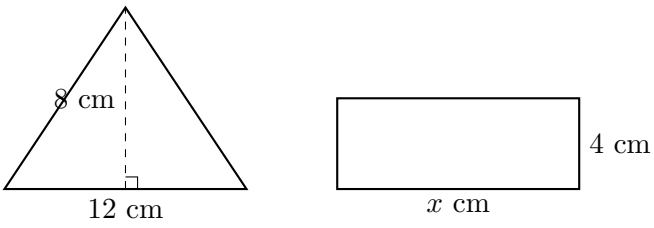
Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

**(d)** Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... $cm^2$ **[2]**

**7** An isosceles triangle and a rectangle are shown below.
The area of the triangle is equal to the area of the rectangle.




NOT TO SCALE

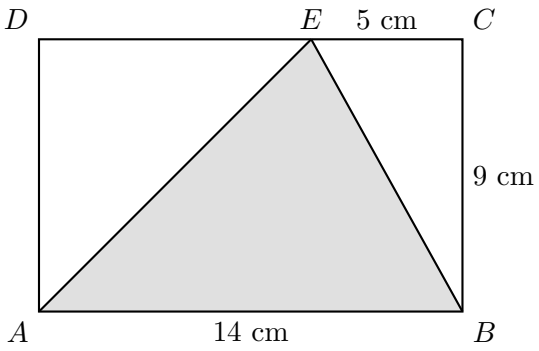
**(a)** Calculate the value of $x$.

$x =$ ..... **[2]**

**(b)** Calculate the perimeter of the isosceles triangle.

..... cm **[3]**

**8** $ABCD$ is a rectangle with length 14 cm and width 9 cm.
The point $E$ lies on the edge $CD$ such that $CE = 5$ cm.



NOT TO SCALE

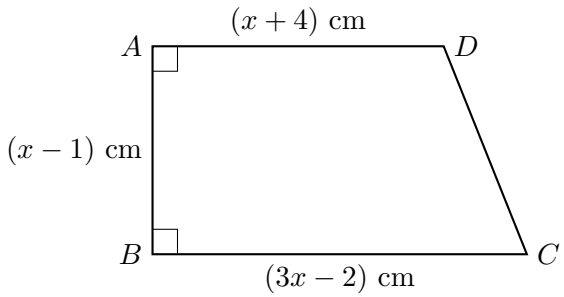
**(a)** Calculate the area of the shaded triangle $ABE$.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the shaded triangle $ABE$.

..... cm **[4]**

**9** A trapezium $ABCD$ has right angles at $A$ and $B$.
The side $AD$ is parallel to $BC$.
$AD = (x + 4)$ cm, $BC = (3x - 2)$ cm and $AB = (x - 1)$ cm.




NOT TO SCALE

**(a)** The area of the trapezium is 20 cm$^2$.

**(i)** Show that $2x^2 - x - 21 = 0$.

**[3]**

**(ii)** Calculate the perimeter of the trapezium.

..... cm **[5]**

**1** **(a)** $2(2x+3)+2(x-1)=34$ OR $2x+3+x-1=17$ **[M1]**

$6x+4=34$ OR $3x+2=17$ **[M1]**

$x=5$ **[A1]**

**(b)** Substitution of $x=5$: $(2(5)+3)\times(5-1)$ OR $13\times 4$ **[M1]**

52 **[A1]**

[Back to Question] *[Total Marks: 5]*

**2** **(a)** $\frac{1}{2}(10+16)\times 8$ **[M1]**

104 **[A1]**

**(b)** Base of right-angled triangle $=16-10=6$ **[M1]**

Slanted edge $=\sqrt{6^2+8^2}=10$ **[M1]**

$8+10+16+10=44$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**3** **(a)** $\frac{1}{2}\times(15+24)\times 8$ **[M1]**

156 **[A1]**

**(b)** Horizontal distance from $D$ to the altitude from $A=\sqrt{10^2-8^2}=6$ **[M1]**

Horizontal distance for the right-angled triangle under $B=24-15-6=3$ **[M1]**

$BC=\sqrt{8^2+3^2}$ $(=\sqrt{73}\approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Back to Question] *[Total Marks: 6]*

**4** **(a)** Area of rectangle $=4(2x+3)$ OR $8x+12$ **[M1]**

Area of triangle $=\frac{1}{2}\times 6\times(3x-1)$ OR $9x-3$ **[M1]**

$8x+12=9x-3$ **[M1]**

$x=15$ **[A1]**

**(b)** Length $=2(15)+3=33$ **[M1]**

Perimeter $=2(4+33)=74$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**5** **(a)** $\frac{1}{2}(x+x+4)(x-1)=40$ **[M1]**

$\frac{1}{2}(2x+4)(x-1)=40$ OR $(x+2)(x-1)=40$ **[M1]**

$x^2+2x-x-2=40 \implies x^2+x-42=0$ **[A1]**

**(b)** $(x+7)(x-6)=0$ **[M1]**

6 **[A1]**

**(c)** Identifies lengths are Top = 6, Bottom = 10, Height = 5 **[M1]**

Horizontal difference = 4, so slanted side = $\sqrt{5^2+4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter = $5+6+10+\sqrt{41}$ **[M1]**

27.4 (or 27.40...) **[A1]**

[Back to Question] *[Total Marks: 9]*

**6** **(a)** Perimeter of rectangle = $6y$ OR Perimeter of square = $60-6y$ **[M1]**

$\frac{60-6y}{4}$ OR $15-1.5y$ **[A1]**

**(b)** $(15-1.5y)^2+y(2y)=113$ **[M1]**

$225-45y+2.25y^2+2y^2=113$ **[M1]**

$4.25y^2-45y+112=0$ leading to $17y^2-180y+448=0$ with no errors seen **[A1]**

**(c)** $\frac{-(-180)\pm\sqrt{(-180)^2-4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180\pm\sqrt{1936}}{34}$ OR $\frac{180\pm44}{34}$ **[M1]**

$y=4$ and $y=6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

**(d)** Identifies $y=4$ as the valid solution (since $15-1.5(6.59)<6.59$) leading to side length = 9 **[M1]**

81 **[A1]**

[Back to Question] *[Total Marks: 10]*

**7** **(a)** Calculates area of triangle: $\frac{1}{2}\times12\times8=48$ **[M1]**

$4x=48$

x = 12[A1]

**(b)** Uses Pythagoras to find slanted side: $\sqrt{6^2+8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter = $12+10+10=32$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**8** **(a)** Uses Area = $\frac{1}{2}\times$ base $\times$ height = $\frac{1}{2}\times14\times9$ **[M1]**

63 **[A1]**

**(b)** Finds $AE=\sqrt{9^2+9^2}=\sqrt{162}$ or 12.7... **[M1]**

Finds $BE=\sqrt{5^2+9^2}=\sqrt{106}$ or 10.2... **[M1]**

Perimeter $= 14 + \sqrt{162} + \sqrt{106}$ **[M1]**

37.0 (or 37.02 . . . ) **[A1]**

[Back to Question] *[Total Marks: 6]*

**9 (a) (i)** Sets up equation for area: $\frac{1}{2}((x+4)+(3x-2))(x-1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x+2)(x-1) = 20$ or $(2x+1)(x-1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

**(ii)** Factorises $(2x-7)(x+3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side $CD$: $\sqrt{(8.5-7.5)^2 + 2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2 + 2.5^2} = \sqrt{7.25}$ (or 2.69 . . . ) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69\cdots = 21.2$ (or 21.19 . . . ) **[A1]**

[Back to Question] *[Total Marks: 8]*